

Course Code:B23EC3101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B. Tech. I Semester MODEL QUESTION PAPER					
ANALOG AND DIGITAL IC APPLICATIONS					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Draw the Pin diagram of Op-Amp.	1	2	2
	b).	Define CMRR.	1	3	2
	c).	Draw the circuit diagram of first order HPF using op-amp	2	2	2
	d).	Draw the circuit diagram of current to voltage converter using op-amp	2	1	2
	e).	Draw the Pin diagram of 555 timer.	3	1	2
	f).	Mention any 4 applications of 555 Timer as Monostable multivibrator.	3	2	2
	g).	What are the different types of DACs?	4	2	2
	h).	List important specifications of ADC.	4	1	2
	i).	Draw the circuit diagram of Decoder using 74x138.	5	2	2
	j).	Draw the Pin diagram of IC 7476.	5	1	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Draw a block diagram of typical OP-AMP and explain the function of each block.	1	3	5
	b).	Explain the operation of Square wave generator circuit with neat circuit diagram and derive expression for time period.	1	4	5
OR					
3.	a).	Explain the operation of a Regenerative comparator with circuit diagram and Waveforms.	1	2	4
	b).	Explain the operation of any 2 of the following op amp applications. (i) Differentiator (ii) Summing amplifier (iii) Logarithmic amplifier	1	4	6
UNIT-2					
4.	a).	Derive the transfer function of a second order LPF. Comment on its frequency response.	2	3	5
	b).	Draw the basic circuit of an op amp based RC phase shift oscillator and explain its operation. Also, derive the expression for frequency of	2	4	5

		oscillation.			
		OR			
5.	a).	Design voltage to current converter using op-amp and then explain its operation.	2	3	5
	b).	Draw the op amp based bandpass filter circuit and briefly explain about its operating principle along with its frequency response curve.	2	3	5
		UNIT-3			
6.	a).	Draw the circuit of Schmitt trigger using IC555 timer and explain its operation?	3	3	5
	b).	Draw and explain the working of 555-timer circuit in astable mode to get output waveform with 50% duty cycle.	3	4	5
		OR			
7.	a).	List important specifications of 566 VCO IC.	3	3	5
	b).	With the help of schematic diagram of 555 timer, explain how it can be used as mono stablemultivibrator	3	4	5
		UNIT-4			
8.	a).	With a neat diagram explain the working principle of R-2R ladder type DAC.	4	3	5
	b).	With a neat block diagram, explain successive approximation type A/D converters in detail.	4	3	5
		OR			
9.	a).	Which is the fastest ADC? Explain the operation and discuss its merits and demerits.	4	3	5
	b).	Draw and explain the circuit diagram of parallel comparator type ADC.	4	3	5
		UNIT-5			
10.	a).	Design an asynchronous Decade counter using IC 7476 and explain the operation.	5	4	5
	b).	Explain the following Sequential logic circuits with suitable ICs. (a) Multiplexer. (b) Priority Encoder.	5	3	5
		OR			
11.	a).	Explain the following Combinational logic circuits with suitable ICs. (a) D flip-flop. (b) JK Flip-flop.	5	3	5
	b).	Design an Universal shift Register using IC74X194 and explain the operation.	5	4	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EC3102					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. I Semester MODEL QUESTION PAPER					
ANTENNAS & WAVE PROPAGATION					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define Half Power Beam Width.	1	2	2
	b).	What is meant by Polarization? And its types.	1	1	2
	c).	List out the salient features of antenna array?	2	2	2
	d).	What is meant by array factor?	2	1	2
	e).	What are the advantages of microstrip antenna?	3	1	2
	f).	What is Spill over and back lobe radiation	3	1	2
	g).	What are the possible errors in antenna measurements?	4	2	2
	h).	What are the methods to measure Phase of an antenna?	4	1	2
	i).	Define wave tilt	5	1	2
	j).	Find the range of LOS system when they receive and transmit antenna heights are 10m and 100m respectively.	5	3	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Derive expressions for the EM fields radiated by a $\lambda/2$ Dipole.	1	3	5
	b).	Discuss the Radiation characteristics of dipole antennas	1	2	5
		OR			
3.	a).	Compute the radiation resistance of a Quarter wave Monopole.	1	3	5
	b).	Explain about different types of current distribution on linear Antennas.	1	2	5
		UNIT-2			
4	a)	Obtain expressions for BWFN and HPBW in Broad-side array.	2	3	5
	b)	Show that in a Uniform linear array, the first side lobe is down the principal maximum by 13.5 db	2	3	5
		OR			
5.	a)	Derive the Electric field radiated by a two element uniform linear array.	2	3	5
	b)	Explain the technique of pattern multiplication with examples	2	2	5

		UNIT-3			
6.	a).	With a neat diagram, explain the operating principles of Log periodic Antenna. List out the disadvantages of a Log periodic Antenna	3	2	5
	b).	Obtain Design parameters of Yagi Uda antenna to operate at 30 MHz.	3	4	5
		OR			
7.	a).	Explain in detail the operating principles of Helical antenna.	3	2	5
	b).	Derive an expression for the expression of the impedance of the Slot Antenna.	3	3	5
		UNIT-4			
8.	a).	Explain in detail about the slotted line method of antenna input Impedance measurement.	4	2	5
	b).	Explain the method of measuring the radiation pattern of an antenna.	4	2	5
		OR			
9.	a).	Explain the measurement of antenna gain using two antenna method.	4	2	5
	b).	Explain the method of measurement of phase of an antenna.	4	2	5
		UNIT-5			
10.	a).	Derive an expression for the refractive index of the Ionosphere.	5	2	5
	b).	Explain ground wave propagation in detail.	5	2	5
		OR			
11.	a).	Derive an expression for the field strength of a Space wave.	5	3	5
	b).	Explain the terms Critical frequency, MUF and Skip distance.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EC3103					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. I Semester MODEL QUESTION PAPER					
DIGITAL COMMUNICATIONS					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	State Nyquist Sampling theorem.	1	1	2
	b).	What is companding?	1	1	2
	c).	What is bit rate?	2	1	2
	d).	Draw the phasor diagram for the QPSK system.	2	1	2
	e).	Define Noise bandwidth.	3	1	2
	f).	Show that the coefficients of spectral components a_k and b_k are uncorrelated to each other.	3	2	2
	g).	What is a matched filter?	4	1	2
	h).	What is a correlator?	4	1	2
	i).	List any two advantages of spread spectrum modulation.	5	1	2
	j).	List any two applications of spread spectrum modulation.	5	1	2
5 x 10 = 50 Marks					
		UNIT-I			
2.	a).	Explain how the linear delta modulation can be used for baseband signal transmission. What are the limitations and how can they be eliminated?	1	2	5
	b).	Explain the operation of Differential Pulse Code Modulation (DPCM) with neat block diagrams.	1	2	5
		OR			
3.	a).	Explain the operation of a PCM system with the help of a neat block diagram and an example.	1	2	5
	b).	With the help of an example, explain the operation of a synchronous TDM-PCM system(T1-Digital System).	1	2	5
		UNIT-II			
4.	a).	Explain the operation of Binary Phase Shift Keying(BPSK) system with the help of neat block diagrams.	2	2	5
	b).	Explain the operation of Minimum Shift Keying (MSK) with neat block diagrams. Sketch the MSK waveforms for the given bit stream $b(t)=$	2	2	5

		0110100 for $m = 5$.			
		OR			
5.	a).	Explain the operation of Quadrature Phase Shift Keying(QPSK) system with the help of neat block diagrams.	2	2	5
	b).	In a DEPSK Receiver, the received bit sequence $b(t)$ is 01101100 then i) Find reconstructed bit sequence $d(t)$ ii) Due to the presence of noise $b(t)$ is recovered as 01111100 then, detect $d(t)$ and also identify the bits which are wrongly detected. Use EX-OR logic.	2	3	5
		UNIT-III			
6.	a).	Explain about linear filtering and calculate noise power output of RC low pass filter and an integrator.	3	2	5
	b).	Explain about frequency domain representation of noise.	3	2	5
		OR			
7.	a).	Explain some sources of noise and narrow band representation of noise.	3	2	5
	b).	What is the effect of filtering on power spectral density of noise and obtain the relation between $H(f)$, input noise PSD $G_{ni(f)}$ & output noise PSD $G_{no(f)}$	3	2	5
		UNIT-IV			
8.	a).	Explain the function of a baseband signal receiver and derive its probability of error?	4	2	5
	b).	By deriving expression for P_e for BPSK and BFSK systems, compare the performance of these two data transmission systems	4	3	5
		OR			
9.	a).	Derive an expression of Probability of error P_e for an optimum filter.	4	3	5
	b).	Derive the expression of Probability of error P_e for a matched filter.	4	3	5
		UNIT-V			
10.	a).	Derive an expression for output SNR when the binary signal is transmitted using BPSK in a PCM system.	5	3	6
	b).	Compare the noise performance of PCM and DM systems.	5	3	4
		OR			
11.	a).	How a CDMA system uses DS Spread Spectrum to provide multiple access communication.	5	2	5
	b).	Explain the operation of Frequency Hopping Spread Spectrum with the help of a neat block diagram.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

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Course Code: B23EC3104					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. I Semester MODEL QUESTION PAPER					
CMOS Digital Integrated Circuits Analysis and Design					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What are the key components of a MOS structure?	1	1	2
	b).	Differentiate between full-custom and semi-custom VLSI design.	1	2	2
	c).	What is the purpose of oxidation in MOSFET fabrication?	2	2	2
	d).	Draw the basic layout structure of a NAND2 gate.	2	3	2
	e).	What is propagation delay in a digital circuit?	3	1	2
	f).	What is short-circuit power dissipation in a CMOS inverter?	3	2	2
	g).	Why are sequential circuits used in digital design?	4	2	2
	h).	What is the function of a D-latch?	4	1	2
	i).	How does volatile memory differ from non-volatile memory?	5	2	2
	j).	What is the role of the capacitor in a DRAM cell?	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the structure and working principle of a Metal-Oxide-Semiconductor (MOS) capacitor.	1	2	6
	b).	Discuss the small-geometry effects in MOSFETs and their influence on circuit behavior.	1	3	4
		OR			
3.	a).	Describe the different steps involved in the VLSI design flow with a suitable block diagram.	1	3	6
	b).	Compare the advantages and disadvantages of resistive-load inverters.	1	4	4
		UNIT-2			
4.		Describe the fabrication process flow of CMOS technology with necessary diagrams.	2	3	10
		OR			
5.	a).	Design and explain the layout of a 2-input NAND gate using CMOS technology.	2	3	6
	b).	Draw a layout for the CMOS Inverter.	2	3	4

		UNIT-3			
6.	a).	Discuss the factors affecting the switching speed of MOS circuits.	3	4	5
	b).	Define propagation delay, rise time, and fall time in digital circuits. Explain their significance.	3	2	5
		OR			
7.	a).	Describe different methods for estimating interconnect parasitics in VLSI design.	3	3	5
	b).	Explain different types of power dissipation in CMOS inverters.	3	2	5
		UNIT-4			
8.	a).	What are bi-stable elements? Explain their behavior with state diagrams.	4	2	5
	b).	What are the limitations of an SR latch, and how are they overcome?	4	4	5
		OR			
9.	a).	Describe the working of a CMOS D-latch with circuit diagram and timing waveforms.	4	3	5
	b).	What is time borrowing in sequential circuits? Explain with an example.	4	3	5
		UNIT-5			
10.		Describe the working principle of an SRAM cell with a circuit diagram.	5	3	10
		OR			
11.		Explain the basic structure and working of a DRAM cell with a suitable diagram.	5	3	10
CO-COURSE OUTCOME			KL-KNOWLEDGE LEVEL		M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EC3105					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. I Semester MODEL QUESTION PAPER					
ELECTRONICMEASUREMENTS ANDINSTRUMENTATION					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is meant by accuracy and precision in measurement?	1	2	2
	b).	Define and explain types of static errors.	1	2	2
	c).	What is a transducer? Give two examples.	2	2	2
	d).	Define active and passive transducers with examples.	2	2	2
	e).	What is an oscilloscope? Mention its basic purpose.	3	2	2
	f).	Define trigger pulse and sweep in CRO.	3	2	2
	g).	What is the purpose of using a Wheatstone bridge?	4	2	2
	h).	Write the applications of capacitance bridges.	4	2	2
	i).	What is a signal generator? Mention its use.	5	2	2
	j).	Define harmonic distortion analyzer.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Define resolution and sensitivity in electronic instruments.	1	3	5
	b).	What is the significance of fidelity and lag in dynamic characteristics?	1	3	5
		OR			
3.	a).	List the types of DC and AC voltmeters and mention one application each.	1	3	5
	b).	Describe the working of a True RMS voltmeter.	1	4	5
		UNIT-2			
4.	a).	What are active and passive transducers? Give one example each.	2	2	5
	b).	State the principle of a piezoelectric transducer.	2	2	5
		OR			
5.	a).	What is a strain gauge? List two types.	2	2	5
	b).	How does a resistance thermometer measure temperature?	2	3	5
		UNIT-3			

6.	a).	Define sweep and trigger pulse in CRO.	3	2	5
	b).	What is a dual trace oscilloscope?	3	2	5
		OR			
7.	a).	List the features of a digital storage oscilloscope.	3	3	5
	b).	How is frequency measured using Lissajous patterns?	3	3	5
		UNIT-4			
8.	a).	What is the purpose of a Wheatstone bridge?	4	2	5
	b).	List the precautions to be taken while using electrical bridges.	4	2	5
		OR			
9.	a).	Define the principle of a Schering bridge.	4	2	5
	b).	Mention one application each for inductance and capacitance bridges.	4	3	5
		UNIT-5			
10.	a).	List different types of signal generators and their outputs.	5	3	5
	b).	What is an arbitrary waveform generator?	5	3	5
		OR			
11.	a).	Differentiate between sine wave and square wave signal generators.	5	3	5
	b).	Define sweep generator and mention its use.	5	4	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

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Estd. 1980

Course Code: B23EC3106					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. I Semester MODEL QUESTION PAPER					
BIO-MEDICAL INSTRUMENTATION					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What are the basic objectives of a medical instrumentation system?	1	2	2
	b).	Define resting and action potentials.	1	2	2
	c).	Define ECG and mention its importance.	2	2	2
	d).	What are the components of the cardiovascular system?	2	2	2
	e).	What is intensive care monitoring?	3	2	2
	f).	List any two instruments used in respiratory system analysis.	3	2	2
	g).	What is bio-telemetry?	4	2	2
	h).	Name any two physiological parameters suitable for bio-telemetry.	4	2	2
	i).	What is the purpose of X-ray instrumentation in medical diagnostics?	5	2	2
	j).	Define electrical shock hazard.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	List the major physiological systems of the human body.	1	2	5
	b).	What is the role of electrodes in biomedical instrumentation?	1	2	5
		OR			
3.	a).	Explain the significance of biomedical signals.	1	2	5
	b).	Mention any two types of bio-potential electrodes.	1	2	5
		UNIT-2			
4.	a).	Explain the working principle of a blood pressure measurement system.	2	3	5
	b).	What is the significance of PQRS and T waves in ECG?	2	3	5
		OR			
5.	a).	Define cardiac output and its measurement method.	2	2	5
	b).	What is plethysmography?	2	2	5
		UNIT-3			

6.	a).	What is the function of a pacemaker?	3	2	5
	b).	What are the elements of patient monitoring equipment?	3	2	5
		OR			
7.	a).	How is respiration analyzed in biomedical systems?	3	3	5
	b).	Explain the need for calibration in patient monitoring systems.	3	3	5
		UNIT-4			
8.	a).	List the main components of a bio-telemetry system.	4	2	5
	b).	What are implantable telemetry units?	4	3	5
		OR			
9.	a).	Mention any two tests carried out on blood in clinical laboratories.	4	2	5
	b).	What is the role of automation in clinical testing?	4	3	5
		UNIT-5			
10.	a).	List the physiological effects of electrical current on the human body.	5	2	5
	b).	What is medical thermography used for?	5	3	5
		OR			
11.	a).	Name any two modern imaging techniques.	5	2	5
	b).	What methods are used for electrical accident prevention in hospitals?	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Estd. 1980

Course Code: B23EC3107					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. I Semester MODEL QUESTION PAPER					
DIGITAL SYSTEM DESIGN					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is the role of a module in Verilog HDL?	1	2	2
	b).	Define gate primitives in Verilog with one example.	1	2	2
	c).	What is a procedural assignment in Verilog behavioral modeling?	2	2	2
	d).	Differentiate between sequential and parallel blocks.	2	2	2
	e).	What is a continuous assignment in dataflow modeling?	3	2	2
	f).	List any two operators used in Verilog dataflow model.	3	2	2
	g).	What is the difference between Moore and Mealy machines?	4	2	2
	h).	Define a user-defined primitive (UDP) with a basic example.	4	2	2
	i).	What is a test bench in Verilog?	5	2	2
	j).	Define design verification.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	List different data types used in Verilog.	1	2	5
	b).	What are system tasks in Verilog? Give examples.	1	2	5
		OR			
3.	a).	Explain the purpose of tri-state gates in digital design.	1	2	5
	b).	What is the use of delay specification in gate-level modeling?	1	2	5
		UNIT-2			
4.	a).	What is the role of loops in behavioral modeling?	2	2	5
	b).	Explain conditional statements used in Verilog.	2	2	5
		OR			
5.	a).	How is a flip-flop designed using behavioral modeling?	2	3	5
	b).	Write a simple Verilog code for a 4-to-1 multiplexer using behavioral model.	2	3	5
		UNIT-3			

6.	a).	How are delays specified in continuous assignments?	3	2	5
	b).	Write a simple Verilog code for a decoder using dataflow modeling.	3	3	5
		OR			
7.	a).	What are the basic transistor switches used in switch-level modeling?	3	2	5
	b).	Differentiate between dataflow and behavioral modeling.	3	3	5
		UNIT-4			
8.	a).	What are the basic components of a state machine?	4	2	5
	b).	Write the advantages of using one-hot encoding in FSM design.	4	2	5
		OR			
9.	a).	What is the purpose of synthesis in digital design?	4	3	5
	b).	Explain the synthesis of sequential logic using flip-flops.	4	3	5
		UNIT-5			
10.	a).	Explain the purpose of assertion verification.	5	2	5
	b).	How are combinational circuits tested using a test bench?	5	2	5
		OR			
11.	a).	Mention two techniques for testing sequential circuits.	5	3	5
	b).	Write a simple Verilog test bench to verify a 2-input AND gate.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Estd. 1980

Course Code: B23EC3108					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. I Semester MODEL QUESTION PAPER					
ARTIFICIAL INTELLIGENCE					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What are AI techniques?	1	2	2
	b).	Define state space search with an example.	1	2	2
	c).	What is the difference between procedural and declarative knowledge?	2	2	2
	d).	Define predicate logic with an example.	2	2	2
	e).	Define non-monotonic reasoning.	3	2	2
	f).	State Bayes' theorem in the context of AI.	3	2	2
	g).	Define semantic nets with an example.	4	2	2
	h).	What is fuzzy logic? Mention one application.	4	2	2
	i).	What is the Mini-Max algorithm in game playing?	5	2	2
	j).	Define natural language processing (NLP).	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the characteristics of a production system.	1	2	5
	b).	What is generate-and-test method?	1	2	5
		OR			
3.	a).	Describe the concept of hill climbing search.	1	3	5
	b).	What is meant by means-ends analysis?	1	3	5
		UNIT-2			
4.	a).	What are Isa and instance relationships?	2	2	5
	b).	Distinguish between procedural and declarative knowledge.	2	3	5
		OR			
5.	a).	Explain simple fact representation in predicate logic.	2	3	5
	b).	What is logic programming in knowledge representation?	2	3	5
		UNIT-3			

6.	a).	Explain the concept of statistical reasoning.	3	2	5
	b).	What are Bayesian networks used for?	3	3	5
		OR			
7.	a).	Define Dempster-Shafer theory.	3	3	5
	b).	Differentiate between rule-based and probabilistic reasoning.	3	3	5
		UNIT-4			
8.	a).	What are frames in AI knowledge representation?	4	2	5
	b).	Explain the concept of conceptual dependency.	4	3	5
		OR			
9.	a).	What are scripts in slot-and-filler structures?	4	2	5
	b).	Compare weak and strong slot-filler structures.	4	3	5
		UNIT-5			
10.	a).	What is alpha-beta cut-off in game search?	5	3	5
	b).	List any two components of a planning system.	5	2	5
		OR			
11.	a).	What is the function of the Hopfield network?	5	3	5
	b).	Distinguish between symbolic AI and connectionist AI.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Estd. 1980

Course Code: B23EC3109					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. I Semester MODEL QUESTION PAPER					
COMPUTER ARCHITECTURE AND ORGANIZATION					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Give an example of a register transfer operation using RTL notation.	1	3	2
	b).	Define a shift microoperation.	1	2	2
	c).	Give an example of an addressing mode.	2	2	2
	d).	What is the role of the timing and control unit?	2	2	2
	e).	Name two modes of data transfer.	3	1	2
	f).	What is the purpose of a priority interrupt?	3	2	2
	g).	List two key features of the 8085 microprocessor.	4	1	2
	h).	What is the primary function of the Accumulator in 8085?	4	2	2
	i).	Name any two 16-bit registers in the 8085 microprocessor.	5	1	2
	j).	Which addressing mode uses a register to directly provide the operand?	5	2	2
5 x 10 =50Marks					
		UNIT-1			
2.	a).	Give the functional organization of a Digital Computer and explain the function of each element of a computer	1	2	5
	b).	Explain one stage of arithmetic logic shift unit with functional table.	1	2	5
		OR			
3.	a).	Design and explain 4-bit arithmetic circuit with functional table.	1	3	5
	b).	Explain about Logical micro operations in detail.	1	2	5
		UNIT-2			
4.	a).	With the help of a diagram explain the functions of the basic computer registers connected to a common bus.	2	2	5
	b).	Explain about various addressing modes with example in detail.	2	2	5
		OR			
5.	a).	Explain about instruction cycle with the help of flow chart.	2	2	5
	b).	What is a stack? Discuss its organization.	2	2	5
		UNIT-3			
6.	a).	Explain Asynchronous data transfer. Discuss about handshaking.	3	2	5

	b).	with a neat diagram, describes DMA transfer in a computer system	3	2	5
		OR			
7.	a).	List and briefly explain various input-output data transfer schemes.	3	2	5
	b).	Explain the memory Hierarchy in the computer system.	3	2	5
		UNIT-4			
8.	a).	Draw the internal architecture of the 8085 microprocessor and explain the functions of its components.	4	2	10
		OR			
9.	a).	List and briefly explain the five status flags in the 8085 flag register.	4	2	5
	b).	Describe the function of the three main buses in the 8085 microprocessor.	4	2	5
		UNIT-5			
10.	a).	Describe the organization of Stack Memory in the 8085.	5	2	5
	b).	Write an 8085 assembly program to multiply 04H by 03H using the repetitive addition method.	5	3	5
		OR			
11.	a).	Explain any three addressing modes of the 8085 microprocessor with one example instruction for each.	5	2	5
	b).	Write an 8085 assembly program to add two 8-bit numbers.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EC3201					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
VLSI DESIGN					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is the relation between I_{DS} and V_{DS} in MOSFET?	1	2	2
	b).	What is latch-up in CMOS circuits?	1	2	2
	c).	What is the purpose of using stick diagrams?	2	2	2
	d).	Draw the layout for the CMOS Inverter?	2	3	2
	e).	Define propagation delay in a CMOS inverter.	3	1	2
	f).	What is scaling and limitations of scaling?	3	1	2
	g).	List two static properties of Complementary CMOS.	4	1	2
	h).	What is the basic principle of Dynamic CMOS logic?	4	2	2
	i).	What is the significance of LUTs in FPGA?	5	2	2
	j).	Define Level Sensitive Scan Design (LSSD).	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the CMOS fabrication steps with neat diagram?	1	3	6
	b).	Compare CMOS, Bi-CMOS and Bipolar Technologies?	1	3	4
		OR			
3.	a).	Illustrate the different forms of pull-up transistors in MOS inverter circuits. Also provide the voltage transfer characteristics for each of them?	1	3	6
	b).	What is latch-up in CMOS circuits? Explain its causes, effects, and prevention techniques?	1	2	4
		UNIT-2			
4.	a).	Draw the Stick Diagram of 2 input NOR gate using CMOS logic?	2	3	6
	b).	Explain about the Lambda based design rules for Transistor?	2	2	4
		OR			
5.	a).	Explain 2 μ m CMOS design rule for wires?	2	2	6
	b).	Explain about double poly CMOS rules?	2	3	4

		UNIT-3			
6.	a).	Explain the concept of sheet resistance (Rs) and how it is applied to MOS transistors and inverters?	3	3	6
	b).	Discuss limitations of scaling. Explain it on the substrate doping?	3	3	4
		OR			
7.	a).	What are inverter delays? Explain the factors that contribute to propagation delay in CMOS inverters?	3	2	6
	b).	Explain about limits due to sub threshold currents?	3	3	4
		UNIT-4			
8.	a).	Explain the working principle of Complementary CMOS logic. What are its key static properties?	4	3	5
	b).	Realize the implementation of two-input NOR gate using CMOS logic. Explain the associated operation by giving truth table?	4	3	5
		OR			
9.	a).	Design and Explain Multiplexer based Latches?	4	2	6
	b).	What is Pass Transistor Logic? Explain how logic gates can be designed using pass transistors?	4	4	4
		UNIT-5			
10.	a).	Explain Basic architecture of FPGA in detail?	5	3	5
	b).	Explain the key features of the Xilinx XC4000 FPGA family. How does it differ from other FPGA families?	5	2	5
		OR			
11.	a).	Explain logical stuck-at-0 or stuck-at-1 faults with the help of suitable examples?	5	3	6
	b).	Explain Built-In-Self Test (BIST) with the help of example?	5	2	4

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EC3202					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
DIGITAL SIGNAL PROCESSING					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	If $X(z)$ is the z-transform of the signal $x(n)$ then what is the ZT of $\text{anu}(n)$?	1	2	2
	b).	Evaluate the Z-transform of $e^{3n}u(n)$?	1	2	2
	c).	What is the ROC of the z-transform of the signal $x(n)=\text{anu}(n)+\text{bnu}(-n-1)$	2	2	2
	d).	What is the DFT of the four-point sequence $x(n)=\{0,1,2,3\}$	2	2	2
	e).	What is the circular convolution of the sequences $X_1(n)=\{2,1,2,1\}$ and $x_2(n)=\{1,2,3,4\}$	3	1	2
	f).	What is the cut-off frequency of the Butterworth filter with a pass band gain $p=-1$ dB at $p=4$ rad/sec and stop band attenuation greater than or equal to 20dB at $s=8$ rad/sec?	3	2	2
	g).	List out the expressions for Hanning and Blackman window functions.	4	2	2
	h).	Illustrate the criteria for selecting window functions used in FIR filter design?	4	1	2
	i).	Which operation has to be performed to increase the sampling rate by an integer factor L ?	5	1	2
	j).	Discuss about anti aliasing filter?	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Find the Z-transform of the signal $x(n)=2n u(n)+3nu(-n-1)$ and its region of convergence.	1	3	5
	b).	Realize the series canonical realization of the following digital transfer function $X(Z) = \frac{z^2 + 2z + 4}{(z - 8)(z^2 - 0.9z + 0.14)}$	1	2	5
		OR			
3.	a).	Compute the response of the following system to the input $u(n)$. Discuss the stability of the given DT system? $y(n)=0.7y(n-1)-0.12y(n-$	1	2	5

		$2)+x(n-1)+x(n-2)$			
	b).	Find inverse Z-Transform of $X(z)=(z^2+2z+3)/(z-1)(z-3)(z-4)$ for i) $ z >4$ ii) $ z <1$	1	4	5
		UNIT-2			
4.		Compute the DFT of the following sequence using Radix-2 DIT FFT algorithm. Show the all intermediate stage results: $x(n)=\{0,1,2,0,2,1,0,2\}$	2	3	10
		OR			
5.		Obtain circular convolution of the two sequences given below using DFT approach $x(n)=\{0,1,0,0\}$, $h(n)=\{2,2,2,2\}$	2	3	10
		UNIT-3			
6.		Design digital Butterworth lowpass IIR filter using BLT method. The filter specifications are given by i) -3dB cutoff frequency at 0.5π rad, ii) at least 15dB attenuation at 0.75π rad	3	4	10
		OR			
7.	a).	Compare Chebyshev and Butterworth analog filters?	3	4	5
	b).	Convert the following analog filter with transfer function using impulse invariance method. $H(s)=s+0.2s+0.2)^2+25$	3	4	5
		UNIT-4			
8.		Design a linear-phase low pass FIR digital filter to meet the following specifications: (i) Pass band = 0 to 10 kHz (ii) Sampling frequency = 100 kHz, $N=11$. Compute the impulse response and transfer function of the desired FIR digital filter using Hamming window?	4	3	10
		OR			
9.	a).	Compare IIR and FIR digital filters?	4	3	5
	b).	Show that FIR filters provide constant group delay and phase delay?	4	4	5
		UNIT-5			
10.		Illustrate the operation of up-sampler, down-sampler, Interpolator and Decimator in time and frequency domains with neat sketches.	5	4	10
		OR			
11.	a).	Explain how Sub band coding of speech signals reduces the bit rate.	5	3	5
	b).	Discuss the effects of finite word length registers.	5	4	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EC3203					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
MICROPROCESSORS AND MICROCONTROLLERS					
Electronics and Communication Engineering					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What are the three groups of signals in 8086?	1	1	2
	b).	What are the advantages of segmented memory?	1	1	2
	c).	Explain the difference between the ADD and ADDC instructions.	2	1	2
	d).	Give the register classification of 8086.	2	1	2
	e).	What is the significance of EA bar pin in 8051?	3	1	2
	f).	Explain PSW register?	3	2	2
	g).	Explain the role of the C/T bit in the TMOD register.	4	2	2
	h).	List the modes of timer in 8051.	4	2	2
	i).	Give different applications of ARM processors.	5	1	2
	j).	What is "Thumb" in ARM processor?	5	1	2
5 x 10 = 50 Marks					
UNIT-1					
2.		Draw and explain the functional block diagram INTEL 8086 Microprocessor.	1	3	10
OR					
3.	a).	Illustrate the generation of a 20-bit physical address in 8086 with an example.	1	3	5
	b).	Draw the flag register of 8086 and explain the function of each flag in detail.	1	2	5
UNIT-2					
4.	a).	Draw the programmable register array of 8086 and explain the function of each Register.	2	2	5
	b).	Write an 8086-assembly language program to find the largest byte for a given block of bytes.	2	4	5
OR					
5.	a).	Explain addressing modes of 8086 with suitable example.	2	3	5
	b).	Write an assembly language program for 8086 to find if given number	2	3	5

		is even or odd.			
		UNIT-3			
6.		Outline the features and explain the internal block diagram of 8051 microcontroller.	3	3	10
		OR			
7.	a).	Explain memory organization in 8051 Microcontroller.	3	2	5
	b).	Classify the instructions of 8051 Microcontroller and Explain basic arithmetic Instructions.	3	2	5
		UNIT-4			
8.	a).	Explain TMOD and TCON register.	4	2	5
	b).	Explain Mode-1 programming of 8051 timer. Describe the different steps to program in Mode-1.	4	3	5
		OR			
9.	a).	With neat diagram write an assembly language program to interface ADC0808 to 8051 microcontrollers.	4	3	5
	b).	Write an assembly language program to generate a square wave on port pin P1.2 of 1kHz using timer-0 in mode 2.	4	3	5
		UNIT-5			
10.	a).	Explain the various ARM families and their features.	5	3	5
	b).	Explain ARM core dataflow model with a neat diagram.	5	3	5
		OR			
11.	a).	With a neat diagram explain the different general purpose registers of ARM processors.	5	2	5
	b).	Explain current program status register (CPSR) with neat diagram.	5	2	5
CO-COURSE OUTCOME			KL-KNOWLEDGE LEVEL		M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EC3204					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
ADVANCED DIGITAL LOGIC DESIGN AND PROTOTYPING ON FPGA					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define a logic gate. Give examples.	1	2	2
	b).	What is the difference between latch and flip-flop?	1	2	2
	c).	Differentiate between Mealy and Moore machines.	2	3	2
	d).	What is state encoding in FSMs?	2	3	2
	e).	What do you mean by "low power FSM"?	3	4	2
	f).	List any two real-time applications of FSMs.	3	4	2
	g).	Define JTAG. What is its role in FPGA programming?	4	2	2
	h).	Mention any two common programming methods used for FPGAs.	4	2	2
	i).	Mention any two examples of vendor-specific IP cores.	5	3	2
	j).	Name any two debugging tools used in FPGA design.	5	3	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the design and operation of basic logic gates using truth tables and Boolean expressions.	1	2	6
	b).	Design a combinational circuit for a 2-bit comparator and explain.	1	2	4
		OR			
3.	a).	Describe the working of an arithmetic logic unit (ALU) with block diagram.	1	2	6
	b).	Explain the operation of SR, JK, D, and T flip-flops with truth tables and logic diagrams.	1	2	4
		UNIT-2			
4.	a).	Explain the structure and working of a Mealy Machine with an example and state diagram.	2	3	6
	b).	What is state encoding? Explain different state encoding techniques (binary, one-hot, gray code) with suitable examples.	2	3	4
		OR			
5.	a).	Design a Moore FSM to detect the sequence "1101". Draw the state diagram and transition table.	2	3	6
	b).	Describe the steps involved in designing a FSM from a given problem	2	3	4

		statement.			
		UNIT-3			
6.	a).	Design a FSM-based vending machine that accepts ₹1 and ₹2 coins and dispenses a product for ₹5. Provide state diagram and Verilog code.	3	4	10
		OR			
7.	a).	What are low power FSMs? Discuss the techniques used to reduce power in FSM design (e.g., clock gating, state encoding).	3	4	6
	b).	Explain the benefits and challenges of implementing FSMs in hardware using HDL (Verilog/VHDL).	3	4	4
		UNIT-4			
8.	a).	Explain the architecture of a typical FPGA. Describe the role of CLBs, I/O blocks, and interconnects.	4	2	5
	b).	Explain the basic components and features of an FPGA development board. Use examples like Xilinx or Intel (Altera) boards.	4	2	5
		OR			
9.	a).	Explain the importance of programming interfaces in FPGA-based digital design. Discuss UART, USB, and JTAG interfaces in this context.	4	2	10
		UNIT-5			
10.	a).	Describe the process of integrating and customizing IP cores in a typical FPGA design flow using vendor tools.	5	3	5
	b).	Compare custom IP cores vs vendor-supplied IP cores in terms of flexibility, optimization, and portability.	5	3	5
		OR			
11.	a).	Describe FPGA prototyping methodologies with emphasis on their advantages for SoC/ASIC design validation.	5	3	6
	b).	Explain the steps for testing and debugging an FPGA-based system using simulation tools and on-board debugging resources.	5	3	4
CO-COURSE OUTCOME			KL-KNOWLEDGE LEVEL		M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EC3205					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
EMBEDDED SYSTEMS WITH ARM CORTEX-M3					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	List two key characteristics of embedded systems.	1	1	2
	b).	Differentiate between Harvard and Von Neumann architectures.	1	2	2
	c).	What is the purpose of special registers in Cortex-M3?	2	2	2
	d).	Explain the significance of the memory map in Cortex-M3 architecture.	2	2	2
	e).	What is the difference between an interrupt and an exception?	3	1	2
	f).	How does NVIC help in reducing interrupt latency?	3	3	2
	g).	Define instruction syntax	4	1	2
	h).	What is the role of CMSIS in embedded system development?	4	2	2
	i).	What is an RTOS, and why is it used in embedded systems?	5	1	2
	j).	What are the different debug modes available in ARM Cortex-M3?	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Describe the RISC architecture and justify its use in ARM-based embedded systems.	1	3	5
	b).	Identify and explain the major challenges faced in embedded system design and development.	1	2	5
		OR			
3.	a).	Why is power efficiency crucial in embedded systems? Suggest techniques to reduce power consumption.	1	2	5
	b).	Describe various real-life applications of embedded systems across different domains.	1	2	5
		UNIT-2			
4.		Describe the architecture of Cortex-M3 with a labeled block diagram. Explain the role of each component	2	2	10
		OR			
5.	a).	Explain the different operation modes of Cortex-M3 and their	2	1	5

		significance in embedded applications.			
	b).	Describe the pipeline mechanism in Cortex-M3. How does it improve instruction execution?	2	2	5
		UNIT-3			
6.		How does Direct Memory Access (DMA) reduce CPU workload? Discuss its advantages.	3	3	10
		OR			
7.	a).	What is the role of the Nested Vectored Interrupt Controller (NVIC)? Explain its features.	3	2	5
	b).	Describe the complete interrupt/exception sequence in Cortex-M3.	3	2	5
		UNIT-4			
8.	a).	Explain the concept of interrupt latency and the factors affecting it.	4	2	5
	b).	Evaluate the benefits of automatic stacking and unstacking during an exception in Cortex-M3.	4	3	5
		OR			
9.		Explain the CMSIS organization with a neat diagram.	4	2	10
		UNIT-5			
10.	a).	Explain the key debugging features of ARM Cortex-M3.	5	2	5
	b).	What is the Trace System in ARM Cortex-M3? Explain its components and working.	5	1	5
		OR			
11.	a).	Discuss how low-power design techniques optimize embedded system performance.	5	2	5
	b).	Explain the working and applications of the SYSTICK timer in ARM Cortex-M3.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EC3206					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
DATA COMMUNICATIONS AND COMPUTER NETWORKS					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	List the components of Data communication system	1	1	2
	b).	Explain the purpose of protocol hierarchies	1	1	2
	c).	What is multiplexing? List the types of multiplexing	2	1	2
	d).	Explain character stuffing and bit stuffing	2	1	2
	e).	Compare Pure ALOHA and Slotted ALOHA	3	1	2
	f).	Define Collision	3	1	2
	g).	List the functions of a repeater	4	1	2
	h).	What is the need for a congestion control algorithm? Name any two	4	2	2
	i).	What is the role of DNS in networking	5	2	2
	j).	What is need of port address in transport layer	5	2	2
5 x 10 =50Marks					
		UNIT-1			
2.	a).	Differentiate the three categories of Networks LAN, MAN and WAN.	1	3	5
	b).	Compare the functions of Network layer and Data link layer of OSI model in transferring data between source and destination separated by multiple networks.	1	3	5
		OR			
3.	a).	Explain different modes of data transmission	1	2	5
	b).	Compare Connection Oriented and Connectionless services	1	3	5
		UNIT-2			
4.	a).	Explain the mechanism of Go Back n and Selective Repeat sliding window protocols	2	2	5
	b).	Explain about various types of transmission media	2	2	5
		OR			
5.		The frame received by the receiver is 10111101100. Using the generator polynomial $G(x) = x^5 + 1$, verify if the frame is correct or damaged.	2	3	5

		UNIT-3			
6.	a).	Explain 802.11 Wifi architecture	3	2	5
	b).	Derive the efficiency of Slotted ALOHA	3	4	5
		OR			
7.	a).	Illustrate the IEEE 802.3 Ethernet standard architecture.	3	3	5
	b).	Explain the concept of Carrier Sense Multiple Access (CSMA) with collision detection.	3	2	5
		UNIT-4			
8.	a).	Distinguish the characteristics of virtual circuit and Datagram subnet.	4	3	5
	b).	b). Differentiate the working of ethernet switch and router.	4	3	5
		OR			
9.	a).	Explain the operation of Leaky Bucket Algorithm. Determine its advantages over the Token Bucket Algorithm.	4	3	5
	b).	Explain about IPV4 Addressing Scheme.	4	2	5
		UNIT-5			
10.	a).	Explain the operation of UDP with the help of its header format. List the applications.	5	3	5
	b).	Explain about elements of Transport protocols.	5	2	5
		OR			
11.	a).	Describe the main function of DNS.	5	2	5
	b).	Explain about the World wide web.	5	2	5
CO-COURSE OUTCOME			KL-KNOWLEDGE LEVEL		M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EC3207					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
RADAR ENGINEERING					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What do you mean by maximum unambiguous range?	1	2	2
	b).	What is called a false alarm?	1	2	2
	c).	what are the functions of the Duplexer	2	2	2
	d).	what are the functions of the Mixer	2	2	2
	e).	Give the function of STALO and COHO in MTI radars	3	4	2
	f).	What is blind speed and why does it occur in MTI Radar?	3	3	2
	g).	What is the significance of the term “monopulse”?	4	2	2
	h).	Distinguish between “Search mode” and “tracking mode” of a Radar.	4	4	2
	i).	What are the advantages of a Phased Array Radar	5	2	2
	j).	How does the repeater jammer work?	5	3	2
5 x 10 =50Marks					
		UNIT-1			
2.		Draw the block diagram of the radar and explain its working	1	3	10
		OR			
3.	a).	Derive the Radar Range Equation.	1	3	6
	b).	What are the System losses of Radar.	1	3	4
		UNIT-2			
4.		Explain in detail about i) Branch type duplexer ii) Balanced type duplexer	2	3	10
		OR			
5.		Explain the various Radar Displays (Scopes)	2	3	10
		UNIT-3			
6.		Illustrate the coherent MTI radar with a neat block diagram and explain the function of each block in detail	3	3	10
		OR			

7.		What is the Doppler effect? A C-band ($f_T = 5000\text{MHz}$) Doppler Radar is to detect all targets with radial velocities greater than 5 miles per hour and less than 60 miles per hour. What are the minimum and maximum Doppler frequencies which Radar must detect?	3	4	10
		UNIT-4			
8.		Draw the block diagram of the amplitude comparison monopulse tracking radar in two coordinates and explain its operation.	4	3	10
		OR			
9.	a).	Draw the block diagram of the Conical scan tracking radar and explain its operation.	4	3	6
	b).	Compare the Sequential lobing and monopulse tracking radar.	4	4	4
		UNIT-5			
10.		Draw the block schematic diagram of a phased array radar and explain its operation.	5	4	10
		OR			
11.	a).	Explain the various techniques that can be used electronically to interfere with Radar performance.	5	4	6
	b).	What are the Electronic counter measures that can be taken to overcome the various techniques that can be used electronically to interfere with Radar performance?	5	4	4

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Estd. 1980

Course Code: B23EC3208					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
MACHINE LEARNING					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Compare Traditional Programming and Machine Learning approaches.	1	2	2
	b).	Distinguish between Quantitative and Qualitative data with examples.	1	2	2
	c).	Give an example for One-Hot-Encoding in feature engineering.	2	3	2
	d).	What is Principal Component Analysis (PCA) used for?	2	2	2
	e).	Give the formula for Linear Regression and explain its components.	3	3	2
	f).	List any four classification algorithms used in Supervised ML.	3	2	2
	g).	What are the differences between Hard and Soft Clustering?	4	2	2
	h).	Define Clustering. Give two real-world applications.	4	1	2
	i).	Mention two advantages of using Cross-Validation.	5	2	2
	j).	List any four evaluation metrics used for regression models.	5	1	2
5 x 10 =50Marks					
		UNIT-1			
2.	a).	Explain the evolution and history of Machine Learning.	1	2	5
		Explain the concept of Machine Learning and how it differs from traditional programming with suitable examples.	1	2	5
		OR			
3.	a).	Describe quantitative and qualitative data types used in ML with real-world examples.	1	3	5
	b).	Explain Learning by Rote and Learning by Induction. Give an example for each to show the difference.	1	3	5
		UNIT-2			
4.	a).	Describe the steps involved in data cleaning and handling missing values and outliers.	2	3	5
	b).	Perform basic Exploratory Data Analysis (EDA) on a given dataset by identifying key patterns or insights and explain how it supports ML model building.	2	3	5
		OR			
5.	a).	Using a sample dataset, apply Univariate, Bivariate, and Multivariate	2	3	5

		analysis and interpret the visualizations or findings from each			
	b).	Given a dataset with categorical variables, perform Feature Engineering using One-Hot-Encoding and explain the resulting transformation.	2	3	5
		UNIT-3			
6.	a).	Apply any one classification algorithm (e.g., Decision Tree or KNN) on a sample dataset and describe the outcome.	3	3	5
	b).	Given a dataset, identify whether it requires classification or regression and justify your choice with model implementation.	3	3	5
		OR			
7.	a).	Use a Random Forest Classifier on a dataset and explain how it improves accuracy compared to a single Decision Tree.	3	3	5
	b).	Train both Decision Tree and Logistic Regression models on a binary classification dataset and compare their confusion matrices.	3	3	5
		UNIT-4			
8.	a).	Given a dataset, apply both Hard Clustering (e.g., K-Means) and Soft Clustering (e.g., Gaussian Mixture Model) and compare the results	4	3	5
	b).	Apply K-Means clustering on a sample dataset and visualize the cluster assignments.	4	4	5
		OR			
9.	a).	Perform Agglomerative Hierarchical Clustering on a given dataset and represent the result using a dendrogram.	4	3	5
	b).	Apply any unsupervised learning technique (e.g., DBSCAN) on a real-world dataset and discuss practical challenges faced during implementation.	4	3	5
		UNIT-5			
10.	a).	Given a confusion matrix for a binary classification problem, analyze how Accuracy, Precision, Recall, and F1-Score are calculated and interpret their significance in evaluating the model.	5	4	5
	b).	Analyze the difference between MSE and RMSE in regression by comparing their impact on model evaluation for datasets with outliers.	5	4	5
		OR			
11.	a).	Analyze the role of Stratified K-Fold Cross-Validation in handling imbalanced classification datasets and compare it with simple K-Fold validation.	5	4	5
	b).	Using a heatmap created from a multivariate dataset, analyze the correlations between features and discuss how this influences feature selection in ML models.	5	4	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code:B23EC3209					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
SMART AND WIRELESS INSTRUMENTATION					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	List any four classifications of sensors.	1	1	2
	b).	Write short notes on the evolution and history of WSN	1	2	2
	c).	Write a short note on the IMote node architecture.	2	2	2
	d).	Mention any two communication interfaces used in node architecture	2	1	2
	e).	What is source encoding	3	1	2
	f).	List different types of modulation used in WSN	3	1	2
	g).	What is Zigbee communication?	4	1	2
	h).	Write two energy management techniques.	4	2	2
	i).	How are seismic events sensed using WSN?	5	2	2
	j).	List any two WSN applications in healthcare.	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the concept of Smart Instrumentation and its evolution with suitable examples	1	3	5
	b).	Describe the communication process in a WSN and its challenges	1	3	5
		OR			
3.	a).	Explain the various design constraints of a WSN — energy, self-management, and security.	1	3	5
	b).	Examine the aspects of decentralized management and wireless networking in a WSN	1	3	5
		UNIT-2			
4.	a).	Explain the architectural overview of a typical node in a WSN.	2	3	5
	b)	Compare microcontrollers, DSPs, ASICs, and FPGAs in node architectures.	2	4	5
		OR			
5	a).	Describe the working of the XYZ node architecture with a neat block	2	3	5

		diagram			
	b).	Discuss SPI and I2C communication interfaces used in WSN nodes.	2	3	5
		UNIT-3			
6.	a).	Explain the basic components of a wireless digital communication system.	3	3	5
	b).	Discuss information transmission over a channel and error recognition/correction.	3	3	5
		OR			
7.	a).	Describe Quadrature Amplitude Modulation with suitable diagrams.	3	3	5
	b).	Explain signal propagation and its challenges in wireless communication systems.	3	3	5
		UNIT-4			
8.	a).	Discuss the development of a WSN based on microcontroller and Zigbee.	4	3	5
	b).	Explain various energy harvesting methods and their principles.	4	3	5
		OR			
9.	a).	Describe different energy management techniques for battery-powered WSN nodes	4	3	5
	b).	Explain how battery selection is calculated for WSN applications.	4	3	5
		UNIT-5			
10.	a).	Explain single and multiple damage detection techniques using natural frequencies and mode shapes.	5	3	5
	b).	Describe precision agriculture and how WSN supports it	5	3	5
		OR			
11.	a).	Discuss the role of WSN in traffic control and healthcare monitoring.	5	3	5
	b).	Explain applications of WSN in underground mining and active volcano monitoring	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EC3211					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
DIGITAL VLSI LAYOUT DESIGN					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Briefly explain the role of masks in VLSI layout.	1	2	2
	b).	What is schematic capturing in the layout workflow?	1	1	2
	c).	Define floor-planning in the context of VLSI layout.	2	1	2
	d).	What is the primary goal of placement in IC layout?	2	1	2
	e).	Distinguish between Design Rule Check (DRC) and Layout vs Schematic.	3	3	2
	f).	Briefly explain the function of Electrical Rule Check (ERC).	3	2	2
	g).	What is a digital cell library?	4	1	2
	h).	Explain the concept of an Euler path in digital cell design.	4	2	2
	i).	Why is it important to consider masks and rules during CAD drawing of an inverter?	5	1	2
	j).	Draw the symbol for a NAND gate and state its boolean expression.	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the complete layout workflow, from schematic capturing to the final drawn layers, highlighting the significance of each stage.	1	2	10
		OR			
3.	a).	Describe the importance of understanding masks and their associated rules in VLSI layout.	1	2	5
	b).	Provide an example of how a mask rule might impact the layout of a simple component.	1	2	5
		UNIT-2			
4.	a).	Define floor-planning in VLSI layout.	2	1	5
	b).	Explain its importance in achieving an optimal design.	2	2	5
		OR			
5.	a).	Explain the concept of routing in VLSI layout.	2	1	5
	b).	Discuss different types of routing techniques used to connect	2	3	5

		components.			
		UNIT-3			
6.	a).	Describe the essential components of a Process Design Kit (PDK).	3	3	5
	b).	Explain their role in facilitating VLSI design and verification.	3	2	5
		OR			
7.	a).	Explain the purpose of Layout versus Schematic (LVS) verification.	3	2	5
	b).	How does it ensure the correctness of a layout?	3	2	5
		UNIT-4			
8.	a).	Explain the Euler path concept.	4	3	5
	b).	how it is applied in the design and routing of digital cells to optimize layout.	4	2	5
		OR			
9.	a).	Discuss the guidelines for standard cell layout.	4	2	5
	b).	How do metal spacing grids influence digital cell design?	4	2	5
		UNIT-5			
10.	a).	Draw the CAD layout of a CMOS Inverter, clearly indicating the different mask layers involved.	5	3	10
		OR			
11.	a).	Draw the CAD layout for an XOR gate, specifying the masks and rules that would be applied in a typical technology.	5	3	10

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EC3212					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
REAL TIME OPERATING SYSTEMS					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Differentiate between hard real-time and soft real-time systems.	1	2	2
	b).	What is a context switch in RTOS?	1	1	2
	c).	Define a semaphore and explain its role in synchronization.	2	2	2
	d).	What are the key differences between binary semaphores and counting semaphores?	2	2	2
	e).	What are the key differences between binary semaphores and counting semaphores?	3	2	2
	f).	Explain the function of an event register in real-time applications.	3	1	2
	g).	What is the difference between an exception and an interrupt ?	4	2	2
	h).	Explain the role of a timer interrupt service routine (ISR) in RTOS.	4	1	2
	i).	What is dynamic memory allocation in embedded systems?	5	1	2
	j).	Define barrier synchronization and explain its significance in RTOS	5	1	2
5 x 10 =50Marks					
		UNIT-1			
2.	a).	Explain the key characteristics of an RTOS that make it suitable for real-time applications.	1	2	5
	b).	Describe the role of a scheduler in RTOS and explain the difference between preemptive priority-based scheduling and round-robin scheduling.	1	2	5
		OR			
3.	a).	Explain the concept of multitasking in RTOS and describe its benefits and challenges.	1	2	5
	b).	Discuss the importance of RTOS objects and services in managing real-time tasks effectively.	1	2	5
		UNIT-2			
4.	a).	Explain the various task states in RTOS and describe how task scheduling is managed.	2	2	5
	b).	Describe briefly the structure of a message queue in RTOS	2	2	5

		OR			
5.	a).	Explain the Typical task operations	2	2	5
	b).	Discuss the in-detail about mutual exclusion (mutex) semaphores	2	2	5
		UNIT-3			
6.	a).	Describe the structure and functionality of a pipe in RTOS. How does it improve data communication between tasks?	3	2	5
	b).	Explain the role of signals in RTOS. Explain signal control blocks	3	2	5
		OR			
7.	a).	Describe the typical operations of pipes	3	2	5
	b).	Discuss event register operations and uses of event registers.	3	2	5
		UNIT-4			
8.	a).	Explain the steps involved in handling a general exception in RTOS.	4	2	5
	b).	Discuss about programmable interval timer	4	2	5
		OR			
9.	a).	Discuss about Programmable Interrupt Controllers and External Interrupts	4	2	5
	b).	Discuss the role of real-time clocks and system clocks in managing time-sensitive tasks in RTOS.	4	2	5
		UNIT-5			
10.	a).	Classify Port-Mapped vs. Memory-Mapped I/O and DMA	5	3	5
	b).	Explain the concept of Fixed-Size Memory Management in Embedded Systems	5	2	5
		OR			
11.	a).	Explain about Hardware Memory Management Units	5	2	5
	b).	Discuss about Resource Synchronization	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EC3213					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
CRYPTOGRAPHY AND NETWORK SECURITY					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is the difference between passive and active security attacks?	1	2	2
	b).	What are the key principles to consider when designing a secure block cipher?	1	1	2
	c).	Differentiate conventional (symmetric) from public key (asymmetric) encryption.	2	2	2
	d).	What are the different modes of operation in DES?	2	1	2
	e).	How does Diffie-Hellman key exchange achieve security?	3	2	2
	f).	What is meant by one-way property in hash function?	3	2	2
	g).	What is the purpose of the Authentication Header (AH) in IPsec?	4	2	2
	h).	How does SSL/TLS ensure secure communication over the web?	4	2	2
	i).	List out the design goals of firewalls.	5	1	2
	j).	What is the role of consensus mechanisms in blockchain?	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Differentiate Active attacks and Passive attacks.	1	2	5
	b).	Explain about Traditional Block cipher Structure.	1	2	5
		OR			
3.	a).	What is mono alphabetic cipher? How it differs from Caesar cipher.	1	2	5
	b).	Explain Block cipher design principles.	1	2	5
		UNIT-2			
4.	a).	Perform Encryption and Decryption using RSA algorithm for $p=17, q=11, e=7, M=88$.	2	3	5
	b).	Explain the structure of AES algorithm with neat diagram and describe the steps in AES encryption.	2	2	5
		OR			
5.	a).	Find the secret key shared between User A and User B using Diffie	2	3	5

		Hellman Key exchange algorithm for the following: $q=97$, $a=5$, the private keys $X_A = 36$, $X_B = 58$.			
	b).	Explain Block Cipher modes of operations.	2	2	5
		UNIT-3			
6.	a).	Illustrate digital signature algorithm with neat diagram and explain how to sign and verify using DSS algorithm.	3	3	5
	b).	Differentiate between HMAC and CMAC.	3	2	5
		OR			
7.	a).	List and explain various steps of SHA in detail with neat diagram.	3	2	5
	b).	Describe Kerberos with steps to grant the ticket.	3	2	5
		UNIT-4			
8.	a).	Describe IP sec architecture with neat diagram.	4	3	5
	b).	Discuss the services provided by PGP with neat diagram.	4	2	5
		OR			
9.	a).	Discuss in detail about SSL/TLS.	4	3	5
	b).	Explain Web security requirements	4	2	5
		UNIT-5			
10.	a).	Explain key elements in Blockchain technology.	5	3	5
	b).	Explain about different types of firewalls.	5	2	5
		OR			
11.	a).	Explain how firewalls are configured.	5	3	5
	b).	Describe how Blockchain technology is used in smart contracts.	5	2	5
CO-COURSE OUTCOME			KL-KNOWLEDGE LEVEL		M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EC3214					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
MICROWAVE ENGINEERING					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	List the microwave applications.	1	2	2
	b).	Explain the probe and Loop.	1	3	2
	c).	List out the scattering properties.	2	2	2
	d).	Explain the significance of scattering matrix.	2	2	2
	e).	Classify microwave tubes.	3	1	2
	f).	Explain the slow wave structures.	3	2	2
	g).	Explain the negative resistance phenomenon.	4	2	2
	h).	What are the applications of solid sate devices?	4	1	2
	i).	List the blocks of microwave bench setup.	5	2	2
	j).	Define the VSWR?	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the operation of a Magic Tee and its applications in detail	1	3	5
	b).	State the principle of operation of Directional coupler. Explain the operation of two hole directional coupler in detail.	1	3	5
		OR			
3.	a).	Explain the operation of circulator	1	4	6
	b).	State Faraday rotation principle. Explain the operation of the isolator.	1	3	4
		UNIT-2			
4.	a).	What is a scattering matrix? Write the properties of a scattering matrix	2	3	5
	b).	Derive the S-parameters for Magic Tee.	2	3	5
		OR			
5.	a).	Show that the 'S' matrix of a perfectly matched 2-port network is $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$	2	3	5
	b).	Explain the operations of directional coupler with the help s-parameters	2	3	5

		UNIT-3			
6.	a).	Explain the limitations of conventional tubes at Microwave frequencies in detail	3	3	5
	b).	Explain the working of Reflex klystron with neat diagram	3	3	5
		OR			
7.	a).	Explain the working of Helix travelling wave tube with neat diagram	3	3	5
	b).	Explain the working of 8-cavity Magnetron with neat diagram	3	3	5
		UNIT-4			
8.	a).	Explain in detail the principle of operation of GUNN diode and detail different modes of operation of gunn diode.	4	3	5
	b).	Explain the operation IMPATT diode with suitable diagrams.	4	3	5
		OR			
9.	a).	Explain the operation TRAPATT diode with suitable diagrams	4	3	5
	b).	Explain the operation TUNNEL diode with suitable diagrams	4	3	5
		UNIT-5			
10.	a).	Explain the procedure with a neat diagram to measure the frequency and guide wave length	5	3	5
	b).	Explain the procedure for measurement of low and high VSWR with block diagram.	5	3	5
		OR			
11.		Draw the Block Diagram of Microwave bench setup and explain each block	5	3	10

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EC3215					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
ANALOG IC DESIGN					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What are the key parameters in a small-signal MOS transistor model?	1	2	2
	b).	Define sub-threshold conduction in a MOSFET.	1	2	2
	c).	What is the purpose of a beta helper in a current mirror circuit?	2	2	2
	d).	List two advantages of using a cascode current mirror.	2	3	2
	e).	What is the role of compensation in operational amplifier design?	3	2	2
	f).	Mention two differences between a differential amplifier and an inverter.	3	3	2
	g).	Define the term hysteresis in the context of comparators.	4	2	2
	h).	What is the key difference between a comparator and an operational amplifier?	4	3	2
	i).	What is a Voltage-Controlled Oscillator (VCO)?	5	2	2
	j).	Name two applications of Phase-Locked Loops.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the significance of passive components in CMOS technology.	1	2	5
	b).	Describe the layout considerations in integrated circuits for MOS design	1	2	5
		OR			
3.	a).	Analyze the small-signal model of a MOS transistor for low-frequency applications.	1	3	6
	b).	Compare different computer simulation models used for MOS device analysis.	1	2	4
		UNIT-2			
4.	a).	Explain the function of a MOS switch and its practical limitations.	2	2	5
	b).	Describe how a current mirror with beta helper improves performance.	2	2	5
		OR			

5.	a).	Analyze the performance of a cascode current mirror in terms of output resistance and compliance voltage	2	3	5
	b).	Explain the principle of operation of a temperature-independent bandgap reference circuit	2	2	5
		UNIT-3			
6.	a).	Describe the operation of a differential amplifier and list its key performance parameters	3	2	5
	b).	Analyze the cascode operational amplifier configuration and explain how it improves power supply rejection ratio (PSRR).	3	3	5
		OR			
7.	a).	Explain the compensation techniques used in two-stage CMOS operational amplifiers.	3	2	5
	b).	Analyze the high gain amplifier architecture and its significance in analog ICs	3	3	5
		UNIT-4			
8.	a).	Define the term comparator and list its important specifications	4	2	5
	b).	Analyze the design improvements to enhance speed and accuracy in open-loop comparators.	4	3	5
		OR			
9.	a).	Differentiate between two-stage and other open-loop comparators.	4	3	5
	b).	Explain how discrete-time comparators operate and where they are used.	4	2	5
		UNIT-5			
10.	a).	Explain the basic principle and frequency calculation of a ring oscillator.	5	2	5
	b).	Compare LC and ring oscillators in terms of phase noise and frequency stability.	5	3	5
		OR			
11.	a).	Describe the operation of a simple PLL and its building blocks.	5	2	6
	b).	Design a basic charge-pump PLL and analyze its non-ideal effects like jitter and lock time.	5	4	4

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EC3216					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
SOFT COMPUTING TECHNIQUES					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define intelligent control and mention any two approaches used in it.	1	1	2
	b).	What is a rule-based system? Give an example.	1	2	2
	c).	Define an Artificial Neural Network (ANN) and mention its basic components.	2	1	2
	d).	What is the McCulloch-Pitts neuron model? State its significance.	2	2	2
	e).	Differentiate between crisp sets and fuzzy sets with an example.	3	2	2
	f).	What are the main steps involved in fuzzy logic control?	3	2	2
	g).	How does the selection process work in a Genetic Algorithm?	4	2	2
	h).	What is the role of crossover and mutation in Genetic Algorithms?	4	2	2
	i).	What are the advantages of using Genetic Algorithms in power system optimization?	5	2	2
	j).	What is the role of the MATLAB Neural Network Toolbox in system identification?	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Describe the architecture of an intelligent control system with a neat diagram.	1	3	5
	b).	What is a symbolic reasoning system? Explain its significance in AI-based control systems.	1	3	5
OR					
3.	a).	Explain the role of knowledge representation in intelligent control with an example.	1	3	5
	b).	Describe how symbolic reasoning systems are applied in intelligent control.	1	3	5
UNIT-2					
4.	a).	Explain the McCulloch-Pitts neuron model with its structure, working principle, and significance in Artificial Neural Networks.	2	2	5
	b).	Explain the basic mathematical model of an Artificial Neural Network	2	3	5

		(ANN) with a suitable example.			
		OR			
5.	a).	Compare Adaline and Madaline networks with respect to their architecture and learning rules.	2	3	5
	b).	Describe the architecture of a Feed-forward Multilayer Perceptron and its applications.	2	3	5
		UNIT-3			
6.	a).	Explain the process of fuzzification and defuzzification in fuzzy logic control.	3	2	5
	b).	Describe how fuzzy modeling is applied for controlling nonlinear systems.	3	3	5
		OR			
7.	a).	Illustrate the structure of a fuzzy logic controller with a block diagram.	3	3	5
	b).	Describe the working of a self-organizing fuzzy logic control system with an example.	3	3	5
		UNIT-4			
8.	a).	Describe the role of selection, crossover, and mutation in Genetic Algorithms.	4	2	5
	b).	Describe the working principle of Tabu Search with an example.	4	3	5
		OR			
9.	a).	How do Genetic Algorithms differ from traditional optimization techniques?	4	2	5
	b).	Illustrate the concept of Ant Colony Optimization (ACO) and its real-world applications.	4	3	5
		UNIT-5			
10.		Analyze the role of Genetic Algorithms in optimizing power system performance. Compare GA-based optimization with conventional techniques and provide a case study to support your analysis.	5	4	10
		OR			
11.	a).	Illustrate the process of implementing a fuzzy logic controller using MATLAB fuzzy-logic toolbox.	5	3	5
	b).	Demonstrate how self-organizing neural networks can be used for system control applications.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks